$q_m = q_1^* q_2$

$\bar{\mathbf{x}} = \frac{1}{n}\sum_i \mathbf{x}_i$

$E_s = \sum_{i \neq j} \|q_i - q_j\|$

κ

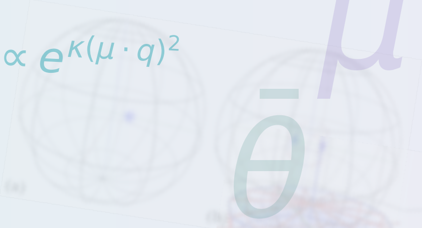
$g(\theta; \mu, \kappa) = \frac{1}{2\pi I_0(\kappa)} e^{\kappa \cos(\theta - \mu)}$

$\bar{C} = \frac{1}{n}\sum_i \cos\theta_i, \ \bar{S} = \frac{1}{n}\sum_i \sin\theta_i$

q_m

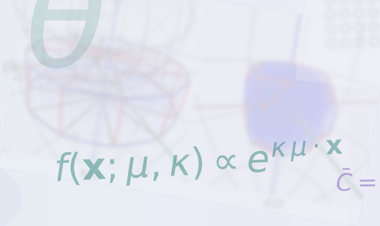
$f_W(q; \mu, \kappa) \propto e^{\kappa(\mu \cdot q)^2}$

μ



$q_m = q_1^* q_2$

$\bar{\theta}$



$w(q; \mu, \kappa) \propto e^{\kappa(\mu \cdot q)^2}$

$E_s = \sum_{i \neq j} \|q_i - q_j\|^{-s}$

$f(\mathbf{x}; \mu, \kappa) \propto e^{\kappa \mu \cdot \mathbf{x}}$

$\bar{C} = \frac{1}{n}\sum_i \cos\theta_i, \ \bar{S} = \frac{1}{n}\sum_i \sin\theta_i$

$\mu, \kappa) = \frac{1}{2\pi I_0(\kappa)} e^{\kappa \cos(\theta - \mu)}$

$f(\mathbf{x}; \mu, \kappa) \propto e^{\kappa \mu \cdot \mathbf{x}}$

$\bar{\mathbf{x}} = \frac{1}{n}\sum_i \mathbf{x}_i$